CBSE MOCK — Pre-Board Practice Series

Central Board of Secondary Education

CLASS XII — CHEMISTRY

Date: 12 November 2026 | Maximum Marks: 70 | Time Allowed: 3 Hours

GENERAL INSTRUCTIONS:

1. This question paper contains 5 sections. Read the instructions for each section carefully.
2. All questions are compulsory unless stated otherwise.
3. Write all answers neatly and show relevant working wherever required.
4. Marks for each question are indicated against it.
5. This paper is a Practice Paper — it is similar to but NOT an actual CBSE board paper.
6. Download more FREE practice papers daily at: **www.cbsemock.org.in**

SECTION A — MCQ & Assertion-Reason (16 Marks)

A/R (1 mark each) | MCQ (1 mark each)

Q1. **[1 mark]**

The van't Hoff factor for a solution of Na_2SO_4 (fully dissociated) is:

(A) 1
(B) 2
(C) 3
(D) 4

Q2. **[1 mark]**

Which of the following is a secondary alcohol?

(A) CH_3OH
(B) C_2H_5OH
(C) $(CH_3)_2CHOH$
(D) $(CH_3)_3COH$

Q3. **[1 mark]**

Which of the following is a natural polymer?

(A) Nylon-6
(B) Bakelite
(C) Starch
(D) Teflon

Q4. **[1 mark]**

Which of the following nucleophiles is most reactive in SN2 reaction?

(A) F^-
(B) Cl^-
(C) Br^-
(D) I^-

Q5. **[1 mark]**

The cell reaction in a Daniel cell is spontaneous because:

(A) Cu^{2+} is a weaker oxidising agent than Zn^{2+}
(B) Zn is a stronger reducing agent than Cu
(C) Zn^{2+} is a stronger reducing agent than Cu^{2+}
(D) Cu is a stronger reducing agent than Zn

Q6. **[1 mark]**

The coordination number of Ni in $[Ni(CO)_4]$ is:

(A) 2
(B) 4
(C) 6
(D) 8

Q7. **[1 mark]**

Which of the following has the highest boiling point?

(A) n-pentane
(B) n-hexane
(C) 2-methylbutane
(D) 2,2-dimethylpropane

Q8. **[1 mark]**

Which colloid is formed when milk is added to water?

(A) emulsion
(B) sol
(C) foam
(D) aerosol

Q9. **[1 mark]**

The shape of $[PtCl_4]^{2-}$ is:

(A) tetrahedral
(B) square planar
(C) octahedral
(D) trigonal bipyramidal

Q10. **[1 mark]**

Half life of a first order reaction is 20 min. The time required for the concentration to drop to 1/8 of initial value is:

(A) 40 min
(B) 60 min
(C) 80 min
(D) 160 min

Q11. **[1 mark]**

Glucose and fructose are related as:

(A) enantiomers
(B) diastereomers
(C) epimers
(D) anomers

Q12. **[1 mark]**

Assertion (A): The melting point of ionic compounds is generally high. Reason (R): Strong electrostatic forces of attraction hold the ions together in an ionic crystal.

Q13. **[1 mark]**

Assertion (A): Ethers are less soluble in water than alcohols of comparable molecular mass. Reason (R): Alcohols can form hydrogen bonds with water while ethers cannot.

Q14. [1 mark]

Which vitamin deficiency causes scurvy?

(A) Vitamin A
(B) Vitamin B_{12}
(C) Vitamin C
(D) Vitamin D

Q15. [1 mark]

In Friedel-Crafts acylation, the electrophile is:

(A) RCOCl
(B) RCO^+
(C) $AlCl_4^-$
(D) $AlCl_3$

Q16. [1 mark]

The antibacterial drug Sulphanilamide belongs to which class?

(A) Antiseptic
(B) Sulfa drug
(C) Antifungal
(D) Antipyretic

SECTION B — Very Short Answer (10 Marks)

VSA (2 marks each)

Q17. [2 marks]

Write the IUPAC name and structure of the compound formed when benzaldehyde undergoes aldol condensation. OR Distinguish between addition polymerization and condensation polymerization with one example each.

Q18. [2 marks]

The standard electrode potential for Zn^{2+}/Zn is -0.76 V and for Cu^{2+}/Cu is $+0.34$ V. Find the cell potential and ΔG° for the Daniel cell.

Q19. [2 marks]

Define 'mole fraction'. Calculate the mole fraction of ethanol in a solution containing 92 g ethanol and 180 g water.

Q20. [2 marks]

Draw the structure of XeF_4 and XeO_3 and predict their shapes.

Q21. [2 marks]

What is meant by 'peptide linkage'? Draw the structure formed when glycine and alanine react to form a dipeptide. OR What are enzymes? Name the enzyme that catalyses the conversion of glucose to ethanol.

SECTION C — Short Answer (21 Marks)

SA (3 marks each)

Q22. [3 marks]

Explain with an example: (a) Electrophilic addition (b) Nucleophilic substitution (c) Free radical mechanism

Q23. **[3 marks]**

(a) Write the chemical reaction involved in the extraction of silver from argentite ore (cyanide process). (b) Why is Zn used in the extraction of silver but not in the extraction of copper? (c) Name one ore of iron and write its formula. OR Explain zone refining. Name the metals refined by this process.

Q24. **[3 marks]**

Explain with examples: (a) Schottky defect (b) Frenkel defect (c) Why does NaCl sometimes appear yellow?

Q25. **[3 marks]**

Write the name and structural formula of the monomer(s) used to prepare: (a) Nylon-6 (b) Bakelite (c) Natural rubber

Q26. **[3 marks]**

The rate constant for a reaction at 500 K is 1.6×10^{-3} mol^{-1} L s^{-1} and at 700 K is 1.6×10^{-2} mol^{-1} L s^{-1}. Find the activation energy using Arrhenius equation. (R = 8.314 J mol^{-1} K^{-1})

Q27. **[3 marks]**

Using VSEPR theory, predict the shapes of: (a) PCl_5 (b) SF_6 (c) ClF_3 Also state the bond angles for each.

Q28. **[3 marks]**

(a) What is tyndall effect? Give an example. (b) Distinguish between lyophilic and lyophobic colloids. (c) What is electrophoresis?

SECTION D — Case Study (8 Marks)

CBQ (4 marks each — Case Study)

Q29. **[4 marks]**

Electrolysis is widely used in the extraction and refining of metals. During electrolytic refining, impure metal is used as anode and pure metal as cathode. The metal dissolves from the anode and deposits at the cathode, while impurities settle as 'anode mud'. Faraday's laws govern the amount of substance deposited: W = ZIt (Z = electrochemical equivalent). (a) State Faraday's first law of electrolysis. (b) Calculate the mass of copper deposited when 2 F of charge is passed through $CuSO_4$ solution. (At. mass Cu = 63.5) (c) Why do impurities not get deposited at the cathode during electrolytic refining? OR (c) How many moles of electrons are transferred when 965 C of charge is passed through an electrolyte?

Q30. **[4 marks]**

Drugs are chemical substances that interact with specific biological targets in the body to produce a biological response. Drugs that bind to and activate receptor proteins are called agonists while those that bind and block are called antagonists. The effectiveness of a drug depends on its ability to bind to the target site (receptor). Aspirin is an antipyretic and analgesic drug that acts by inhibiting an enzyme cyclooxygenase (COX). (a) What is meant by 'receptor'? (b) Distinguish between agonist and antagonist drugs with examples. (c) Why should drugs not be taken without medical advice? OR (c) Name the class of drugs used to treat bacterial infections and give one example.

SECTION E — Long Answer (15 Marks)

LA (5 marks each)

Q31. **[5 marks]**

(a) Describe the following reactions with example: (i) Reimer-Tiemann reaction (ii) Kolbe's reaction (b) Explain the mechanism of SN1 reaction with an example. Why do tertiary halides undergo SN1 more readily? (c) Convert: Bromobenzene to biphenyl (Wurtz-Fittig reaction) OR (a) Write the product(s) of the following reactions: (i) $CH_3CHO + HCN \rightarrow$ (ii) C_6H_5CHO + aniline $\rightarrow$ (iii) $CH_3COCH_3 + NH_2OH \rightarrow$ (b) How is acetic acid prepared from acetaldehyde industrially? Give the chemical equation. (c) Explain Cannizzaro reaction with an example.

Q32. **[5 marks]**

(a) Write the names and formulae of the ores of: (i) Copper (ii) Aluminium (iii) Iron (b) Explain the following processes giving suitable chemical equations: (i) Smelting of copper (ii) Thermite welding (c) Why is aluminium not extracted by carbon reduction? OR (a) Give an account of the metallurgy of copper. (b) Explain the following: (i) Cast iron is brittle but steel is not (ii) Pig iron is not useful industrially

Q33. **[5 marks]**

(a) State Henry's law. How does it explain the solubility of gases in liquids? (b) Calculate the osmotic pressure of a solution obtained by mixing 100 mL of 3.4% solution (w/v) of urea (M = 60) and 100 mL of 1.6% solution (w/v) of sucrose (M = 342) at 27°C. ($R = 0.0821$ L atm K^{-1} mol^{-1}) OR (a) What is elevation of boiling point? Derive the expression relating it to molality of solution. (b) 18 g of glucose (M = 180) is dissolved in 1 kg of water. Calculate the boiling point of the solution. (K_b for water = 0.52 K kg mol^{-1})

✦ END OF QUESTION PAPER ✦

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